

SQL Task

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› Prompt 1: Understanding SQL and Tables

Act as a SQL instructor.

Explain what SQL is and why it is used. Create a simple example table named

`employees` with columns such as:

- `employee_id`
- `name`
- `department`
- `salary`

Explain the purpose of each column and how tables store data.

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› Prompt 2: SELECT and WHERE Clause

Act as a SQL trainer.

Create a sample `employees` table and insert sample records. Demonstrate how to use:

- `SELECT`
- `WHERE`

Show examples to:

- retrieve employees from a specific department
- filter employees with salary greater than a given value

Explain how the `WHERE` clause helps filter data in databases.

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› Prompt 3: ORDER BY and LIMIT

Act as a SQL expert.

Using an example table, demonstrate:

- ORDER BY
- LIMIT

Show how to:

- sort records in ascending and descending order
- retrieve only the top few records

Explain real-world use cases of sorting and limiting results.

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› Prompt 4: Aggregate Functions

Act as a SQL instructor.

Explain aggregate functions such as:

- COUNT
- SUM
- AVG

Use a simple table (such as employees or sales) to demonstrate how these functions summarize data.

Explain why aggregation is useful in data analysis.

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› Prompt 5: GROUP BY and HAVING

Act as a SQL trainer.

Explain the difference between:

- WHERE
- HAVING

Use a sample table to:

- group records using GROUP BY
- filter grouped results using HAVING

Provide simple examples and explanations.

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> Prompt 6: INNER JOIN

Act as a SQL expert.

Create two related tables, such as:

- employees
- departments

Demonstrate an INNER JOIN to combine data from both tables. Explain how joins help retrieve related information.

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> Prompt 7: LEFT JOIN

Act as a SQL instructor.

Explain LEFT JOIN with a simple example. Show how LEFT JOIN returns all records from one table even if matching records do not exist in the other table.

Explain when LEFT JOIN is useful.

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> Prompt 8: Handling Duplicate Records

Act as a SQL analyst.

Explain how duplicate records occur in databases. Demonstrate how to:

- identify duplicates
- remove duplicates using DISTINCT

Explain why handling duplicates is important.

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> Prompt 9: Subqueries

Act as a SQL expert.

Explain what subqueries are. Provide a simple example where a subquery is used inside a SELECT or WHERE clause.

Explain the logic step by step.

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› Prompt 10: SQL Concept Summary

Act as a SQL instructor.

Summarize the most important SQL concepts covered:

- SELECT
- WHERE
- GROUP BY
- HAVING
- JOINS
- SUBQUERIES

Explain how these concepts work together in real-world databases.

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